BUSINESS INTELLIGENCE I | 2026-2027

How could data improve animal welfare and shelter operations?

Austin Animal Center | Initial data analysis

173,812 intakes	173,775 outcomes	2013-2025
Recorded arrivals	Recorded outcomes	Historical observation window

Overall assessment
A useful source for understanding demand, outcomes and time in care. It supports a credible shelter operations project, but encounter matching, linked sampling, primitive measures and five meaningful hierarchies still need validation.

What the evidence suggests

Plan for changing demand. Animal type, intake reasons and seasonal patterns can guide staffing, foster outreach and adoption activity.

Make long waits visible. Among 168,787 provisional matched encounters, the median recorded interval is 6 days and the mean is 21.0. A single average hides very different experiences.

Protect decisions from misleading data. Repeat visits, missing outcome subtypes and mixed timestamp representations require explicit treatment before performance comparisons.

How to read the report

Raw-event charts use every supplied record, retaining duplicate candidates. Stay-duration figures use an explicitly restricted, deduplicated matching analysis. Proposed operational actions are hypotheses to evaluate, not measured welfare improvements.

These historical exports end in May 2025. They do not describe current shelter conditions in October 2026.

01 / DATA FOUNDATION

Two event streams; one animal can have several visits

Source	One row represents	What it can establish
Intakes	One recorded intake event	Arrival demand, intake circumstances and animal characteristics
Outcomes	One recorded outcome per animal per encounter [S3]	Recorded departure classifications and animal characteristics at outcome

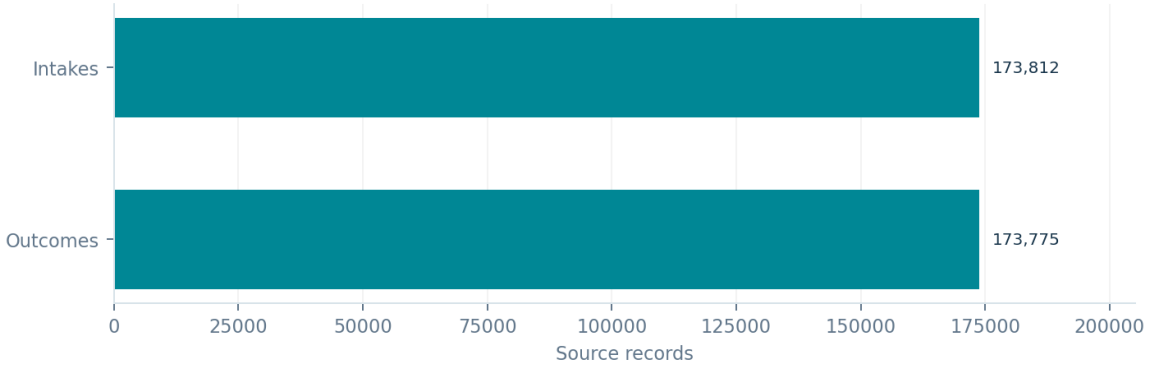


Figure 1. Full-file source record counts. Both files exceed the assignment cap of 30,000 rows per project source dataset; preserve originals and prepare documented linked extracts.

Evidence	Intakes	Outcomes
Distinct animals	156,287	156,237
Animals with multiple records	13,487	13,507
Exact duplicate excess rows	36	26
First event	2013-10-01	2013-10-01
Last event	2025-05-04	2025-05-05

155,423 Animal IDs appear in both files; 864 appear only in intakes and 814 only in outcomes. Missing counterparts may reflect observation boundaries or record gaps; they are not automatically open shelter stays.

01A / WORKED RELATIONSHIP EXAMPLE

Connect encounters, rather than every row for an animal

The following actual records belong to Animal ID **A245945**. Names are omitted because they are unnecessary for the relationship example.

Event	Recorded date and time	Category
Intake	07/03/2014 05:55:00 PM	Stray
Intake	05/20/2015 10:34:00 PM	Stray
Outcome	2014-07-04T15:26:00	Return to Owner
Outcome	2015-05-25T11:49:00	Transfer

Two intake rows and two outcome rows create four combinations when joined only on Animal ID. Only a chronological, validated encounter rule can establish which pairs are defensible.

Full-file join warning
A direct Animal ID inner join would produce 221,246 row combinations. This is a count of combinations, not a count of verified shelter encounters.

Proposed warehouse pattern

Shared dimensions	Separate facts	Encounter analysis
Animal and Date, plus event-specific intake/outcome classifications	One intake event per row; one outcome event per row	A separate reconstructed encounter fact, with match status and source lineage

Aggregate separate event facts to a common, justified level before comparing them. A shared animal key or date dimension does not make every intake/outcome combination a valid stay.

This is a proposed model, not an implemented Fabric warehouse. Animal descriptors can change between encounters; reproductive status and age must remain event-specific.

02 / SHELTER DEMAND

Understand who arrives and why

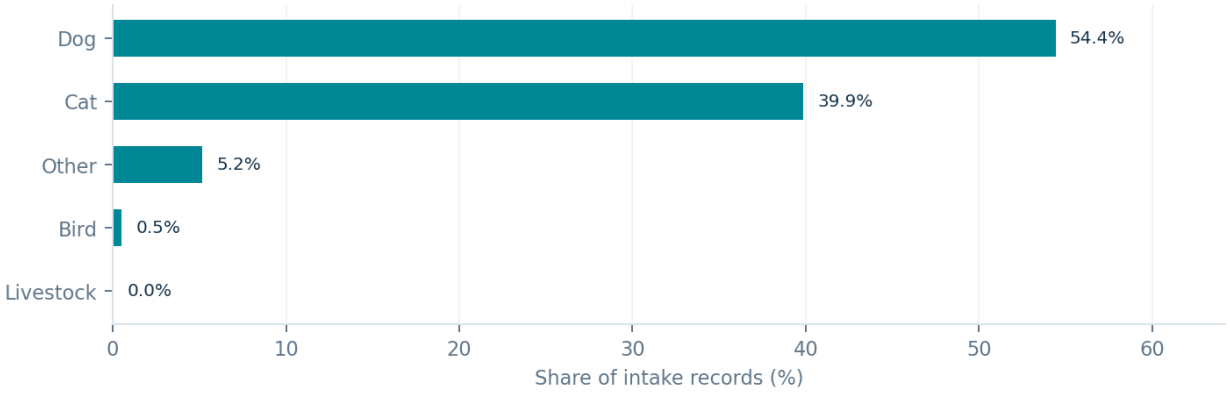


Figure 2. Animal type as a share of all 173,812 intake records. These are event shares, not shares of unique animals.

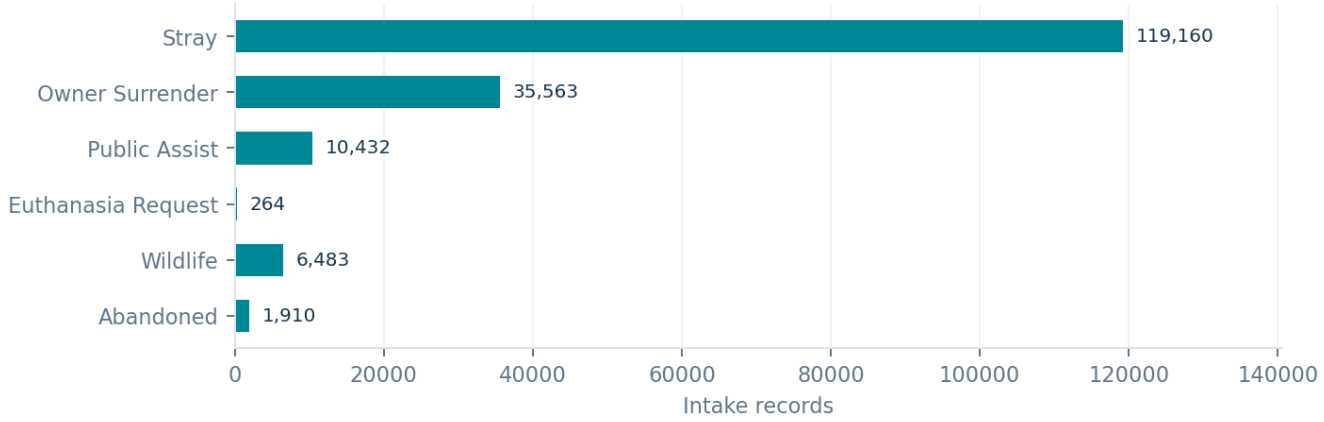


Figure 3. All intake reasons in the supplied file. Stray arrivals and owner surrenders describe different operational demands; they should not be treated as one undifferentiated workload.

Proposed operational use

Use animal type and intake reason to plan staffing, foster outreach and reunion activity. Protect rare groups in sampling: only 34 intake records are Livestock. Event counts do not measure care complexity, kennel capacity or staff hours.

03 / TIME AND SEASONALITY

Plan for peaks, while respecting the observation window

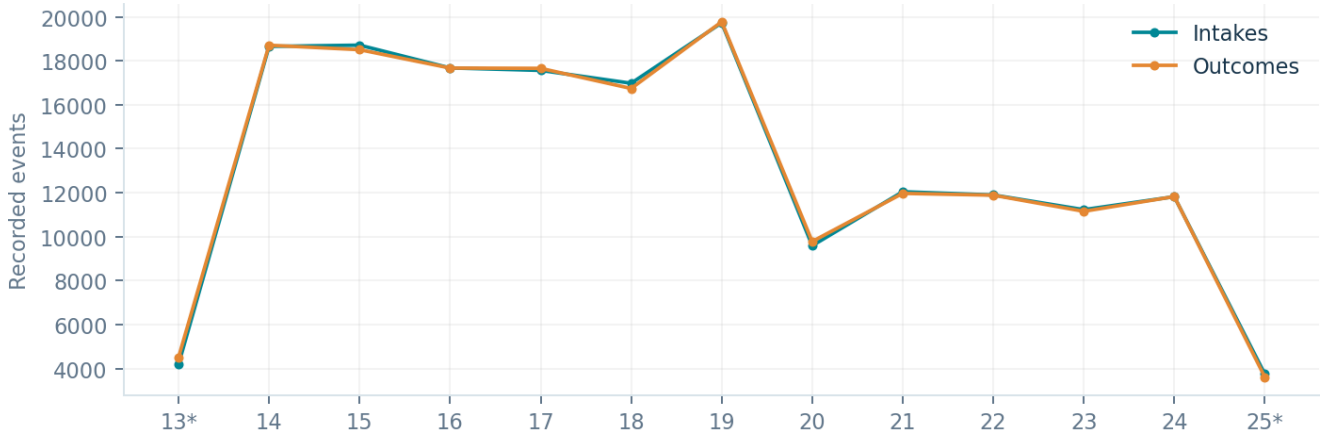


Figure 4. Annual raw-event counts; * marks partial years (2013 starts in October; 2025 ends in May). Intake and outcome volumes are not an occupancy calculation.

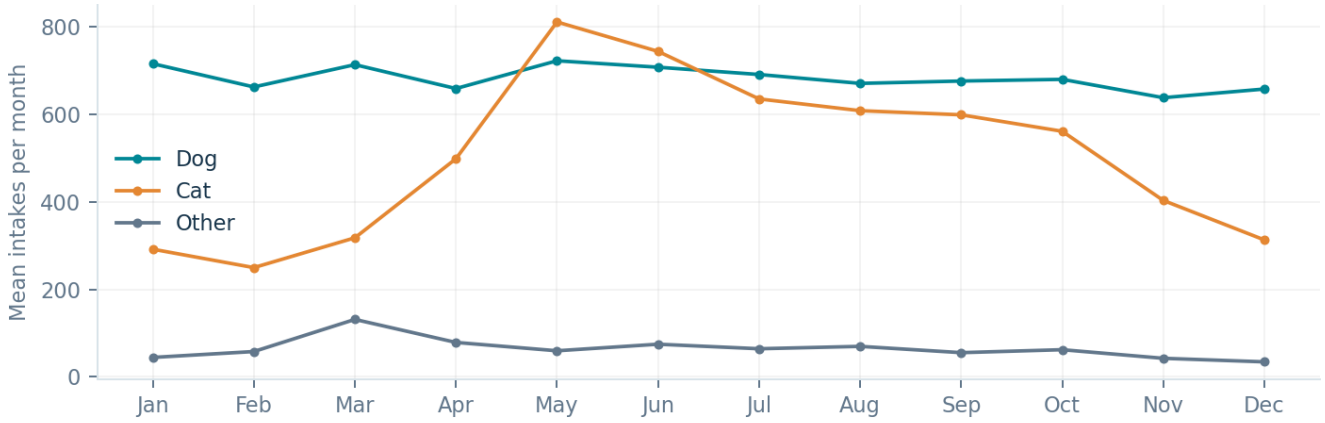


Figure 5. Mean intakes for each calendar month across the 11 complete years 2014-2024, separately for Dogs, Cats and Other. Bird and Livestock are excluded from this chart only.

The historical average Cat intake peak occurs in **May** (811 recorded intakes per such month). Use this as an input to seasonal planning, then compare recent local data before changing resources.

Year-to-year changes may reflect policy, reporting, access or demand changes. This descriptive chart does not establish their causes or forecast future volumes.

04 / RECORDED OUTCOMES

An outcome mix is more informative than one headline rate

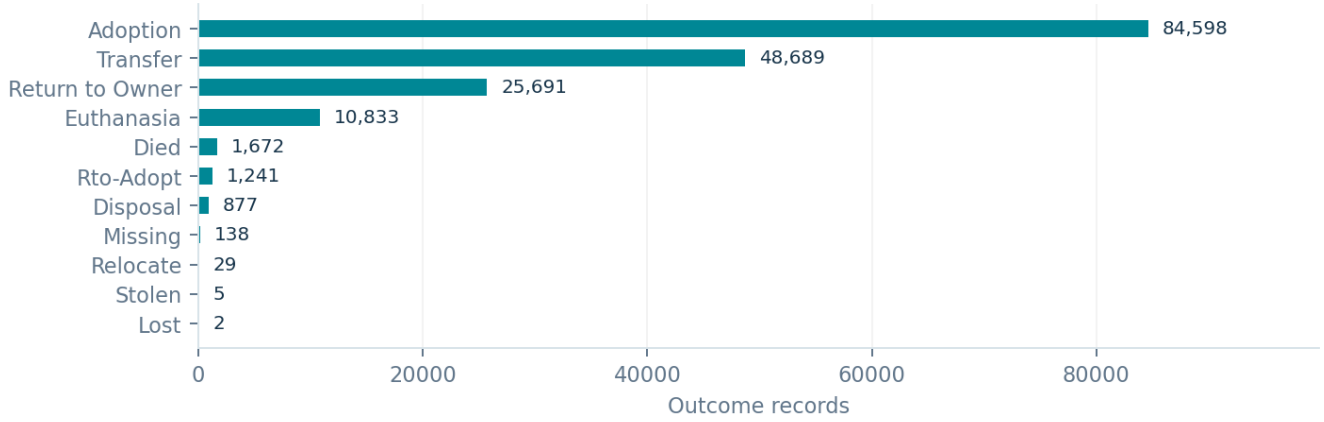


Figure 6. All 173,775 raw outcome records. Adoption: 84,598 (48.7%). The Missing bar combines 92 records labelled Missing and 46 blank outcome types; all remain in the denominator.

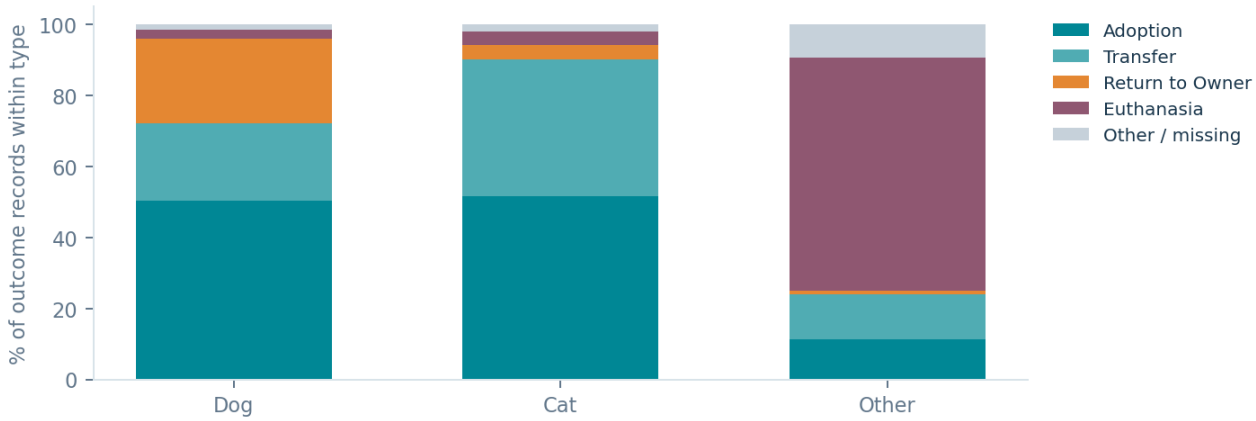


Figure 7. Within-type outcome shares for Dogs, Cats and Other; Other / missing combines all remaining outcome types and blanks. Bird and Livestock are omitted from this comparison. Outcome subtypes are not used to define these groups.

A lower duration or higher adoption share alone does not demonstrate better welfare. Transfers are not verified final adoptions; outcome categories also reflect case mix and organizational practice. Review the underlying groups and denominators before setting targets.

05 / PROVISIONAL TIME IN CARE

The long tail changes the planning problem

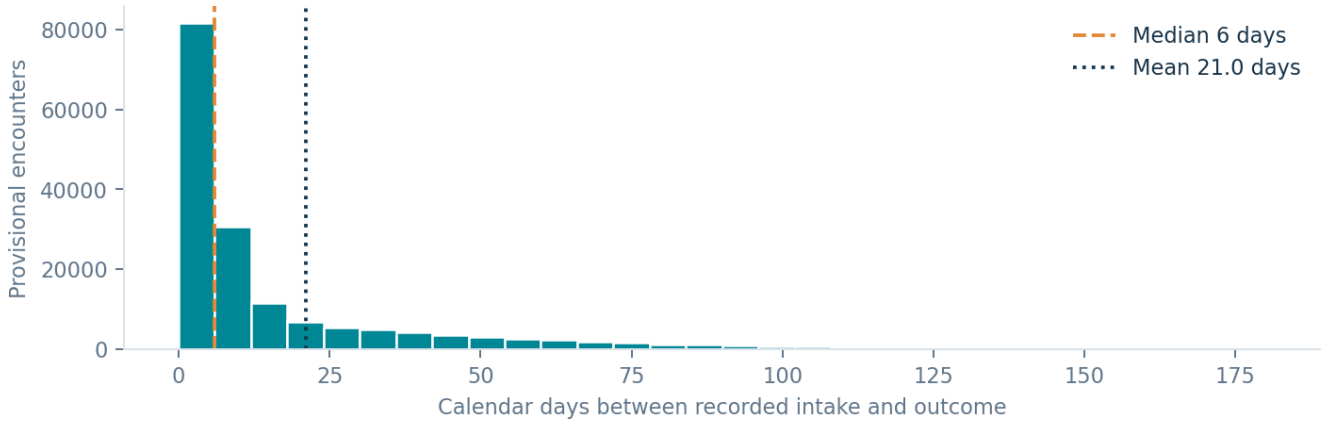


Figure 8. Calendar-day intervals for provisional matched encounters, displayed through 180 days. 2,307 longer intervals are outside the chart window but remain in every overall statistic.

Matched n	Mean	Median	90th percentile	Range
168,787	21.0 days	6 days	56 days	0-1913 days

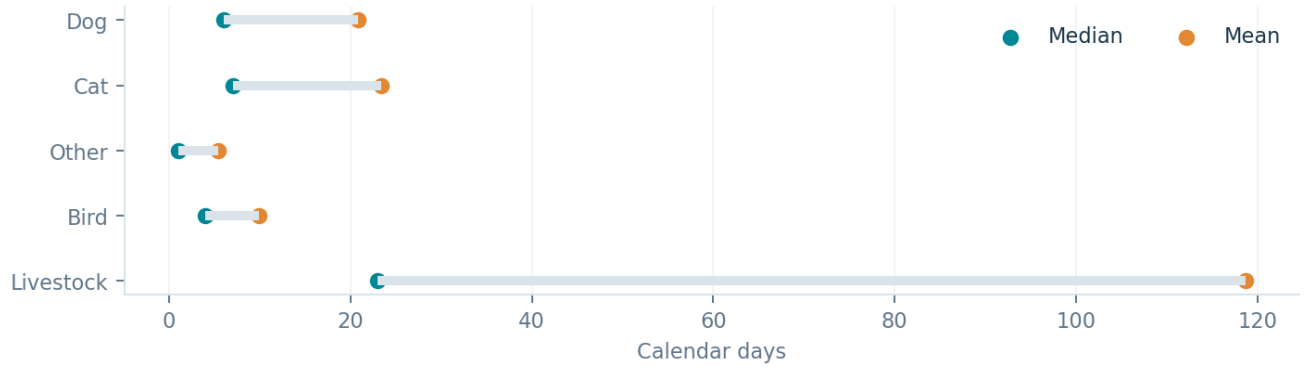


Figure 9. Mean versus median provisional intervals by intake animal type. The Livestock estimate has only 30 matched encounters; it should not drive a broad comparison.

Proposed welfare use

32,038 provisional intervals exceed 30 days (19.0%). A live case-review queue could help staff investigate long waits and barriers. Thirty days is illustrative, not a welfare threshold. These historical completed matches exclude unresolved cases.

05A / MATCHING INTEGRITY

Duration estimates need an auditable encounter rule

Step	Rule actually used in this exploration
1	Remove exact duplicate rows within each source for the matching analysis only.
2	For each animal, sort events by recorded wall-clock timestamp.
3	Consider outcomes at or after an intake and strictly before its next intake.
4	Accept only one candidate outcome, with no tied intake timestamp.
5	Exclude explicit-offset outcome timestamps and identical intake/outcome times.
6	Calculate outcome date minus intake date, in calendar days; prevent outcome reuse.

Intake classification after deduplication	Count
Provisional match	168,787
Explicit offset excluded	3,249
No outcome in interval	1,632
Multiple outcomes in interval	77
Tied intake timestamp	12
Same timestamp excluded	19

The rule produces 168,787 provisional matches from 173,776 deduplicated intake rows (97.1%). A match is an algorithmic inference, not a source-supplied encounter ID.

Calendar days are not elapsed 24-hour periods. Excluding offset-bearing candidates avoids mixing timestamp representations in accepted pairs, but remaining timezone semantics still need confirmation. Missing events, incorrect dates and re-entry sequences can cause plausible-looking false matches. The matched population is selected and may underrepresent difficult cases.

06 / COMPLETENESS AND CLEANSING

Missing values require field-specific policies

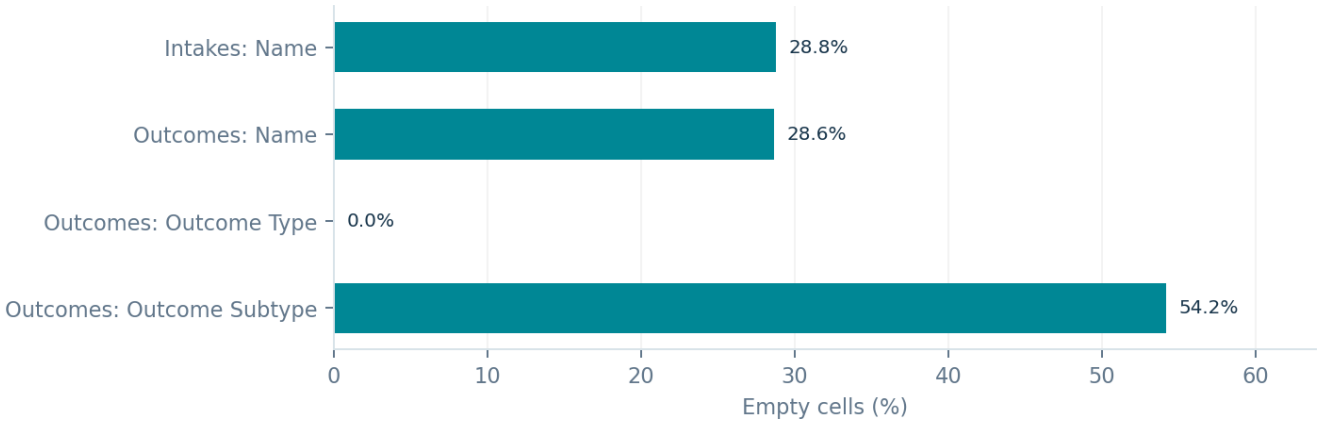


Figure 10. Empty CSV cells as a percentage of all records in each source. Literal NULL values are counted separately, not silently converted during profiling.

Finding	Meaning / proposed treatment
Outcome subtype: 94,115 empty (54.2%)	A subtype may be inapplicable or unavailable. Do not invent one or drop all such rows.
Outcome type: 46 empty	Retain an explicit unknown classification and log for review.
Names: 49,991 / 49,784 empty	Names are optional descriptors, not business keys.
Sex: one literal NULL in each file	Normalize explicit unknowns using a documented policy.
36 intake / 26 outcome exact duplicate excess rows	Investigate source identity before final removal; preserve lineage.
Animal ID: no empty cells	Useful linkage key, but not a unique encounter key.

Operational implication

A reliable review list must distinguish an unknown outcome from a verified open case. Missing names should not make animals disappear from reporting. Use quality flags and a review workflow rather than blanket imputation.

07 / AGE, CONDITION AND EXTREMES

Use unusual values to ask better questions

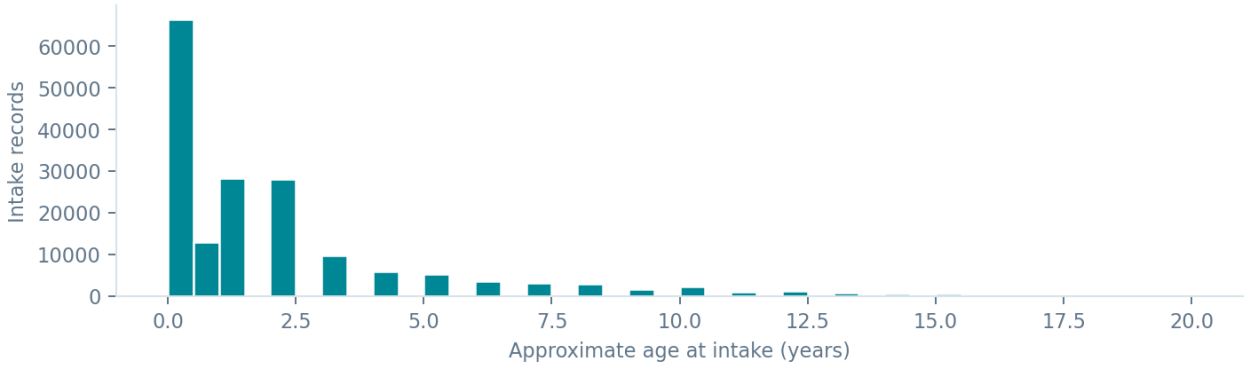


Figure 11. Approximate age at intake, displayed up to 20 years; 13 older records are outside the visual window but retained in statistics. Recorded units are standardized using 365.25 days/year and 365.25/12 days/month; invalid negative values are excluded. Rounded source ages are not exact ages.

Numeric feature	Mean	Median	Min / max	IQR flags
Intake age (years)	1.94	1.00	0.00 / 30.00	22,585
Outcome age (years)	1.98	1.00	0.00 / 30.00	23,001
Provisional interval (days)	21.00	6.00	0 / 1913	20,648

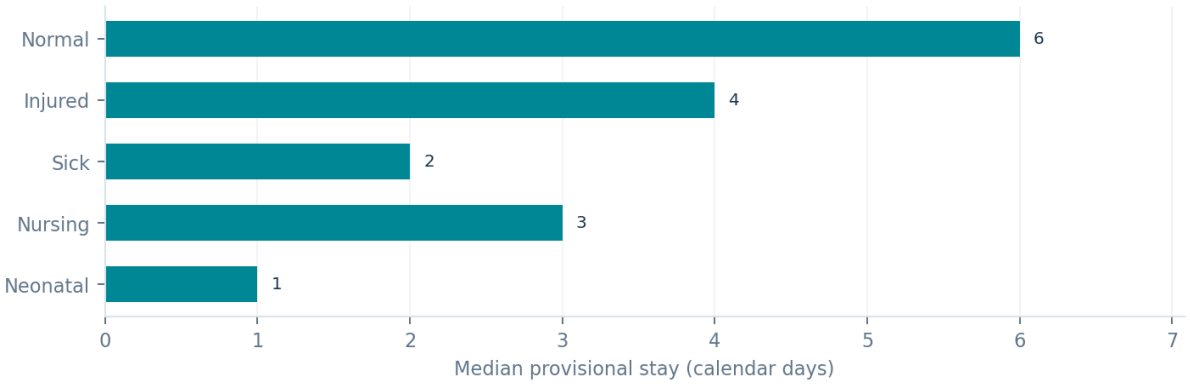


Figure 12. Median provisional interval for five selected intake conditions; see numerical reference for group sizes and tail statistics. Shorter intervals for vulnerable groups do not demonstrate recovery or better care.

Negative age strings occur in 10 intake rows and 10 outcome rows; investigate rather than interpreting them as real ages. IQR flags identify statistical extremes (outside Q1 - 1.5 x IQR or Q3 + 1.5 x IQR), not errors. Long stays may be valid and important to retain.

08 / FROM EVIDENCE TO ACTION

Measure better experiences, not just faster throughput

Data signal	Proposed action	How to evaluate it
Seasonal animal-type demand	Schedule foster recruitment and staff coverage ahead of recurring peaks.	Compare peak-period demand with service capacity; add staff/foster availability data.
Long or repeated recorded stays	Create a staff-reviewed long-wait queue with reasons and follow-up.	Track reviews, barriers resolved and later outcomes; avoid speed-only targets.
Stray intake and return-to-owner patterns	Assess reunion outreach and identification workflows.	Track documented reunion attempts and time to reunion; add attempt records.
Outcomes differ across groups	Review adoption and transfer pathways for comparable groups.	Compare within age/type/condition cohorts; check follow-up outcomes where available.
Repeat intakes	Review recurring cases and preventive support options.	Define re-entry windows and follow-up completeness; measure recurrence carefully.
Unknown or ambiguous records	Introduce quality review and correction logging.	Monitor unresolved records, correction turnaround and reconciliation counts.

What the supplied data cannot measure directly

Pain, stress, enrichment, treatment quality, kennel crowding, staffing levels, foster availability and post-adoption stability are not observed here. Add responsibly collected operational records if these become project objectives. Do not label a duration chart a direct measure of welfare.

A practical project aim
Help shelter managers identify demand peaks, long waits and data gaps, then evaluate whether targeted actions improve animal experiences and operational reliability. This report identifies opportunities; it does not demonstrate an intervention effect.

09 / BI ASSIGNMENT READINESS

A feasible project, with two modelling gaps to resolve

Requirement [S1]	Current assessment	Next evidence
More than one year of history	Meets date-span requirement.	Choose a multi-year analysis window.
Source >=10k fact rows; DW >=5k	Source volumes are ample; final valid fact count pending.	Count after matching, sampling and exclusions.
Each source dataset <=30k rows	Both full exports exceed the cap.	Document linked extracts; preserve originals.
At least 5 dimensions incl. Date	Date, Animal, Classification, IntakeReason, IntakeCondition, Outcome and AgeBand are candidates.	Define grain, keys and event-specific attributes.
5 hierarchies averaging 3 levels	Not established. Date is straightforward; others need mappings.	Validate meaningful roll-ups; obtain guidance.
2 distinct primitive measures	Encounter count and duration proposed; acceptance unconfirmed.	Confirm teacher interpretation; supplement if required.
Fabric staging and warehouse	Not implemented or verified by this report.	Separate staging database and independent STG/DW pipelines.
Copy Data + Dataflows; DQ LOG	Exploration provides candidate checks only.	Use both tools and four exact lab-defined checks.

Candidate hierarchy mappings include animal type / breed group / breed, outcome family / type / subtype, and age classifications. Their depth alone does not prove analytical validity. Free-text locations may not support a reliable geographic hierarchy.

No teacher readiness score, schema approval, Fabric load, successful ETL or delivery is asserted. The assignment guide remains the source for formal requirements.

10 / PROPOSED LINKED SAMPLING

Keep encounters interpretable while reducing source size

Candidate extract	Observed feasibility count	Status
Intakes dated 2022-2023	23,123 raw rows; 23,114 after exact deduplication	Candidate two-year cohort
Animals referenced by that cohort	21,914 distinct IDs	Candidate parent-key set
All deduplicated outcomes for those animals across full history	24,415 rows	Feasibility count; includes events outside cohort encounters

These counts show that a linked approach may fit the 30,000-row limit without random sampling. No extract has been generated. The broad outcome count is not the final matched-outcome sample and must not be used as a cohort outcome denominator.

Proposed sequence

1. Confirm the business questions, primitive measures and encounter rules.
2. Select the multi-year intake cohort and retain stable source row identifiers.
3. Keep full-history events available during validation to detect prior visits and subsequent intake boundaries.
4. Export only the justified cohort events, required outcome follow-up and dimension keys, with a documented exclusion policy.
5. Reconcile source counts, category coverage, unmatched cases and final fact volume; save the sampling manifest.

Boundary rule
An animal admitted late in 2023 can have an outcome in 2024 or 2025. Include a defined follow-up window and flag unresolved observations at its end. Do not quietly discard slow or unresolved cases to obtain a cleaner duration distribution.

Keep originals in data/raw and place derived extracts in data/processed. A selected cohort represents its selected population; it does not establish total shelter occupancy. Current occupancy would require complete valid event histories and appropriate opening balances.

APPENDIX A / FIELD REFERENCE

All 24 source fields: completeness and distinct values

Full raw files. Missing = empty cell; distinct counts exclude empty cells. IDs, names, categories and dates have no meaningful arithmetic mean or median.

Intakes

Field	Empty %	Distinct nonempty
Animal ID	0.00	156,287
Name	28.76	29,774
DateTime	0.00	119,722
MonthYear	0.00	140
Found Location	0.00	70,183
Intake Type	0.00	6
Intake Condition	0.00	20
Animal Type	0.00	5
Sex upon Intake	0.00	6
Age upon Intake	0.00	55
Breed	0.00	3,006
Color	0.00	661

Outcomes

Field	Empty %	Distinct nonempty
Animal ID	0.00	156,237
Date of Birth	0.00	8,692
Name	28.65	29,756
DateTime	0.00	146,382
MonthYear	0.00	140
Outcome Type	0.03	11
Outcome Subtype	54.16	26
Animal Type	0.00	5
Sex upon Outcome	0.00	6
Age upon Outcome	0.00	56
Breed	0.00	3,010
Color	0.00	663

APPENDIX B / NUMERICAL REFERENCE

Typical stays and tails must be considered together

Only provisional matched encounters; calendar-day intervals. These statistics retain every accepted interval, including values beyond the histogram window.

Intake type	n	Mean	Median	P90	Maximum
Dog	92,102	20.8	6.0	50.0	1,913
Cat	67,028	23.4	7.0	63.0	1,745
Other	8,816	5.5	1.0	10.0	681
Bird	811	9.9	4.0	24.0	237
Livestock	30	118.7	23.0	386.6	635

Selected intake conditions

Condition	n	Mean	Median	P90
Normal	143,048	21.1	6.0	54.0
Injured	10,449	23.7	4.0	67.0
Sick	7,689	14.6	2.0	41.0
Nursing	3,753	23.3	3.0	70.0
Neonatal	1,869	19.0	1.0	69.0

Group differences combine case mix, observation limits and matching selection. A short recorded interval can end in different outcomes; it should not be read as a successful recovery. Very small groups, especially Livestock, require particular restraint in interpretation.

Statistical definitions

The median is the 50th percentile; P90 is the 90th percentile. Quartiles and percentiles use linear interpolation (NumPy default). IQR = Q3 - Q1. Duration Q1 = 2 days and Q3 = 20 days, so the upper 1.5 x IQR boundary is 47 days. Values above that boundary are review candidates, not a deletion list.

Overall provisional interval	Value
Matches	168,787
More than 30 days	32,038 (19.0%)
More than 180 days	2,307 (1.4%)
IQR-flagged intervals	20,648 (12.2%)

APPENDIX C / METHODS AND SOURCES

Full-file evidence, with explicit analytical limits

Coverage and reproducibility

All 347,587 records and all 24 source fields were inspected. Raw hashes were checked before and after analysis and matched. Event charts retain raw rows; matching removes only exact row duplicates. No raw file was edited. No sampling was executed.

Empty strings and literal NULL values were counted separately. Dates use their recorded US or ISO representation. Age conversions use 7 days/week, 365.25/12 days/month and 365.25 days/year; 10 negative ages in each file and 9 unparsed outcome ages are excluded from age statistics. Birth-date consistency has not been validated. Date strings and identifiers are not numeric measures.

Matching is described on page 8. Excluded records remain in raw-event charts. Calendar-day intervals describe a selected set of algorithmically paired records; they are not verified physical kennel occupancy, treatment duration or final placement success. No confidence intervals or causal effects are claimed.

Sources and evidence

[S1] Business Intelligence I Project Guide 2026-2027 v1.1, 13 October 2026. Local source: BI Project/docs/source/. Requirements pp.2-6; report pp.7-8; delivery p.9.

[S2] City of Austin, Austin Animal Center Intakes (10/01/2013 to 05/05/2025). Historical catalogue; Public Domain. Consulted 26 October 2026.
https://data.austintexas.gov/d/wter-evkm

[S3] City of Austin, Austin Animal Center Outcomes (10/01/2013 to 05/05/2025). Catalogue defines a row as an outcome per animal per encounter; Public Domain. Consulted 26 October 2026.
https://data.austintexas.gov/d/9t4d-g238

[S4] Supplied local originals: Austin_Animal_Center_Intakes_(10_01_2013_to_05_05_2025)_20261026.csv and Austin_Animal_Center_Outcomes_(10_01_2013_to_05_05_2025)_20261026.csv. Actual event coverage and counts were measured directly. Original download date is unverified.

[S5] User-supplied BI_Project_Initial_Data_Analysis.pdf (15 pages), used only as an analysis, writing and visual reference. Its Airbnb findings were not reused as shelter evidence.

Next incomplete task
Validate the proposed encounter rule against source context and edge cases, settle measures and hierarchies with the teacher, then prepare the linked cohort and Fabric implementation. Welfare and operations actions require current data and measured follow-up before claiming improvement.